Ashif S. Iquebal

Associate Professor, School of Computing and Augmented Intelligence, Arizona State University

aiquebal@asu.edu · (979) 739-2685 · Google Scholar · LinkedIn

Research Summary

My research focuses on causal inference (using counterfactual explanations and generative models such as flow matching), sequential experimentation (active learning, Bayesian optimization), and scientific machine learning (stochastic Bayesian inference, SINDy and its variants) for data-driven decision making in complex systems. My work has been published across 40+ peer-reviewed publications including NAACL, TPAMI, Nature Communications Physics, IEEE Transactions on Signal Processing, and JAIR, and has earned finalist/winner recognition in 8 best paper competitions across IEEE, INFORMS, and IISE. My research has been funded by over $11M from NSF, NIH, DOD, and industry partners. I am fluent in Python (PyTorch, JAX), with hands-on experience in R and scalable statistical tooling.

Education

Ph.D., Industrial & Systems Engineering — 2020
Texas A&M University

M.S., Statistics — 2019
Texas A&M University

B.Tech., Industrial & Systems Engineering and **Minor** in Math and Computing — 2014
Indian Institute of Technology, Kharagpur

Professional Experience

Associate Professor — Fall 2026
Graduate Faculty of Industrial Engineering and Computer Science
School of Computing and Augmented Intelligence, Arizona State University

Assistant Professor — Fall 2020–2026
Graduate Faculty of Industrial Engineering and Computer Science
School of Computing and Augmented Intelligence, Arizona State University

Selected External Funding

Total $11.6M in generative modeling for molecular discovery and materials design, AI-driven decision making, and ML/AI for autonomous systems.

- **NSF** – Accelerating Computational Additive Manufacturing – $500k (2025-28)
- **DOD Army** – Accelerate Materials Design and Process Optimization through AI/ML, $10M (2024–28)
- **DOD MxD** – Executing what if scenarios at hyperscale via virtual manufacturing $150K (2023–25)
- **NIH** – Information-Theoretic Surprise-Driven Approach to Enhance Decision Making, $700K (2023–27)
- **SRP** – Tracking of Spare Parts for Supply Chain Circularity & Waste Characterization, $145K (2023–24)
- **MK Morse** – Active Learning for Process Optimization in Saw Blade Manufacturing, $15K (2024)
- **Mayo Clinic** – Diagnosis and Prediction of Genetic Mutations via -omics and Imaging, $100K (2023–25)

Selected and Recent Publications (Refer to Google Scholar for Complete List)

AI/ML FOR SCIENTIFIC DISCOVERY AND CAUSAL INFERENCE

1. A. Iquebal. Velocity Fields of Optimal Transport Reveal Causality: Towards Root Cause Analysis via Flow Matching. Working Paper, ICLR 2027

2. R. Olabiyi, H. Hu, and A. Iquebal. Discovering Governing Physics in the Presence of Uncertainty. *Communications Physics*, 2026

3. U. J. Islam, K. Paynabar, G. Runger, and A. S. Iquebal. Dynamic Exploration–Exploitation Trade-off in Active Learning with Bayesian Hierarchical Modeling. *IISE Transactions*, 57(4):393–407, 2025

4. S. Zhou, U. Islam, N. Pfeiffer, I. Banerjee, B. K. Patel, and A. S. Iquebal. SCGAN: Sparse CounterGAN for Counterfactual Explanations in Breast Cancer Prediction. *IEEE Transactions on Automation Science and Engineering*, 2023

5. S. Zhou, N. Pfeiffer, U. Islam, I. Banerjee, B. K. Patel, and A. S. Iquebal. Generating Counterfactual Explanations for Causal Inference in Breast Cancer Treatment Response. In *2022 IEEE 18th International Conference on Automation Science and Engineering (CASE)*, pages 955–960. IEEE, 2022

6. A. S. Iquebal, S. Pandagare, and S. T. S. Bukkapatnam. Learning Acoustic Emission Signatures from a Nanoindentation-Based Lithography Process: Towards Rapid Microstructure Characterization. *Tribology International*, 143:106074, 2019

GENERATIVE AI FOR SCIENCES

7. K. Coutinho, L. Dai, and A. Iquebal. Fine-tuning molecular foundation models via adjoint matching for data-scarce discovery. Working Paper, ICML 2027

8. S. Kumbhar, V. Mishra, K. Coutinho, D. Handa, A. Iquebal, and C. Baral. Hypothesis generation for materials discovery and design using goal-driven and constraint-guided LLM agents. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 7524–7555, 2025

9. P. Wu, A. S. Iquebal, and K. Ankit. Emulating Microstructural Evolution During Spinodal Decomposition using a Tensor Decomposed Convolutional and Recurrent Neural Network. *Computational Materials Science*, 224:112187, 2023

10. A. S. Iquebal, P. Wu, S. Ali, and K. Ankit. Emulating the Evolution of Phase Separating Microstructures using Low-Dimensional Tensor Decomposition and Nonlinear Regression. *MRS Bulletin*, 2023

11. P. Wu, W. Farmer, A. S. Iquebal, and K. Ankit. A Novel Data-Driven Emulator for Predicting Electromigration-Mediated Damage in Polycrystalline Interconnects. *Journal of Electronic Materials*, 2023

OTHER AI/ML FOUNDATIONS AND APPLICATIONS

12. S. Zhou, E. Gel, and A. Iquebal. Accelerating Point-Based Value Iteration via Active Sampling of Belief Points and Gaussian Process Regression. *Journal of Artificial Intelligence Research*, 2026. (Accepted)

13. R. Liu, H. Yang, and A. Iquebal. Optimizing self-organizing network for thermal simulation in additive manufacturing. *Journal of Manufacturing and Systems Engineering*, 2026. (Accepted)

14. M. Kothandaraman and A. Iquebal. From Orthographic Freehand Sketches to 3D CAD: A Geometry-Aware Multi-View CAD Primitive Generation Framework. Working Paper, ICLR 2027

15. A. Varghese, B. Starly, and A. Iquebal. Vector Quantization of Complex Sensor Data in Limited Network Bandwidth and Edge Hardware Constrained Digital Twins. *Manufacturing Letters*, 2026. (Accepted)

16. R. Olabiyi, J. Weaver, and A. S. Iquebal. Characterization of Elastoplastic Properties of Additively Manufactured Specimens from Indentation Data using Stochastic Inverse Modeling. Proceedings of the ASME 2024 Manufacturing Science and Engineering, 2024

17. A. S. Iquebal and S. T. S. Bukkapatnam. Consistent Estimation of the Max-Flow Problem: Towards Unsupervised Image Segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(5):2346–2357, 2022

18. Z. Wang, Z. Wang, W.-H. Ko, A. S. Iquebal, V. Nguyen, N. A. Kazerooni, Q. Ma, A. Srinivasa, P. R. Kumar, and S. Bukkapatnam. An Autonomous Laser Kirigami Method With Low-Cost Real-Time Vision-Based Surface Deformation Feedback System. *The International Journal of Advanced Manufacturing Technology*, 118(5):1873–1883, 2022

19. A. S. Iquebal, S. T. S. Bukkapatnam, and A. Srinivasa. Change Detection in Complex Dynamical Systems using Intrinsic Phase and Amplitude Synchronization. *IEEE Transactions on Signal Processing*, 68:4743–4756, 2020

Honors and Awards

- Trailblazer Award for New and Early Stage Investigators, NIBIB/NIH (2023); NSF Manufacturing Blue Sky Competition Finalist (2022); ASU Top 5% Teaching Award (2022)
- **Best Paper Competition Finalist/Winner**:
  - IISE Manufacturing & Design (2026) — *Optimizing Self-Organizing Network for Thermal Simulation in Additive Manufacturing*
  - INFORMS Quality, Statistics, and Reliability (2024) — *Discovering Governing Physics in the Presence of Uncertainty*
  - IEEE CASE Healthcare Automation (2022) — *SCGAN: Sparse CounterGAN for Counterfactual Explanations in Breast Cancer Prediction*
  - IISE QC&RE (Best Paper, 2019) and INFORMS QSR (Best Poster, 2018) — *Consistent Estimation of the Max-Flow Problem: Towards Unsupervised Image Segmentation*
  - INFORMS Data Mining (2018) — *Consistent Estimation of the Max-Flow Problem: Towards Unsupervised Image Segmentation*
  - INFORMS Data Mining (2016) — *Change Detection in Complex Dynamical Systems using Intrinsic Phase and Amplitude Synchronization*
- Third Place, IISE Pritsker Doctoral Dissertation Award (2021)
- **Other notable awards**: NSF Travel Grants (2018, 2016), Fellow – Academy of Future Faculty, Texas A&M (2018), Graduate Teaching Fellowship (2017), Runner-up – ASA Southeastern Texas Best Student Poster (2017), INFORMS Doctoral Colloquium Travel Grant (2017)

Mentoring & Academic Impact

- **Mentoring.** Advising/advised 7 PhD students and committee member of 15 PhD thesis, 9 MS students, and over 50 undergraduate students
- **Teaching.** Developed and taught Advanced Quality Control, Mathematical Statistics with an average instructor evaluation of 4.6/5
- **Service.** Council/Board member for INFORMS QSR (2023–present), NAMRC Scientific Committee (2024–present), and IISE Manufacturing & Design (2024–present) and Data Analytics (2022–24). Track chair, MSEC 2023. Reviewer for 15+ journals and conferences including IISE Transactions, IEEE TASE, IEEE TNNLS, Additive Manufacturing, and Journal of Quality Technology